

Smart Controller System using RTOS

1. Objective:

Develop a Real-Time Operating System (RTOS) based Smart Climate Control System that monitors and adjusts the temperature and humidity of an environment using sensors and actuators. The system will control a cooler, heater, and humidifier to maintain optimal conditions. Since this project is a demo for RTOS course, some logical operations are proposed to play a demo, and not be closed to real operations.

2. Components:

Sensors: **DHT20** for temperature and humidity sensing, connected to I2C port (red port). The **LCD** can be added for data visualization

Actuators:

- Heater: device display 3 colors, using RGB LED, connected to D3 and D4
- Cooler: device display 3 colors, using RGB LED, connected to D5 and D6
- Humidifier: device display 3 colors, using RGB LED, connected to D7 and D8

Microcontroller: YoloUNO platform, which is based on ESP32 S3. A recommended devices can be found in the figure below

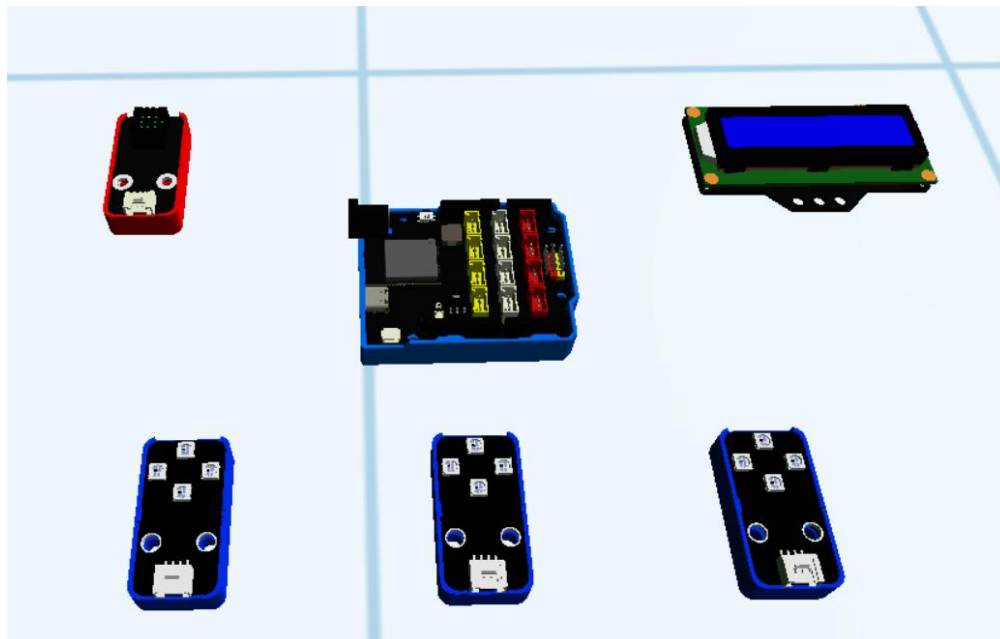


Figure 1: YoloUNO, DHT20, LCD and 3 RGB LEDs

3. RTOS:

FreeRTOS and all concepts are allowed to use in the project.

PLEASE USE THE SEMAPHORE for TASK COMMUNICATIONS

4. Features:

Monitoring:

- Read temperature and humidity from sensors and update on Serial Monitor every 5 seconds.
- The onboard LED (LED 13) is blink every second to monitor the status of the board.

Control Logic:

- Heater (LED D3 D4): Define 3 thresholds for the cooler, display the **GREEN (LED connected in D3 and D4)** color when temperature in a safe range, **ORANGE** color and **RED** color are for risky ranges. Describe your range in the report.
- Cooler (LED D5 D6): Activate the heater if the temperature is higher (LED is GREEN) a threshold. The cooler is activated for a fix duration (e.g. 5 seconds), then the temperature is checked again.
- Humidifying: Activate the humidifier if humidity falls below a predefined threshold. However, the progress of activating of the humidifier is quite complex. **The LED (connected to D7 and D8) is GREEN for 5 seconds, then YELLOW for 3 second. Finally, it is RED for 2 seconds.** The humidity is checked again, to determine that this progress is repeated or not.

5. Task Management:

Describe your tasks here. Present the implement of your task. Only main source code is required in this part. **If there is a state machine in your task, please show your state machine**

- 5.1 Blinky Task
- 5.2 Read Temperature Task
- 5.3 Cooler Task
- 5.4 Heater Task
- 5.5 Humidifier Task

6. Results:

At the end of this part, show the sharable link of your project. Some snapshots can be presented in this section as well. The submission guidance will release later.